Two platforms, one configuration: independent exact certification of the dimension-eleven kissing records

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Abstract

The kissing number $K(d)$ is the largest number of non-overlapping unit spheres that can touch a central unit sphere in $\mathbb{R}^d$. In dimension eleven the record stood near 582 for decades and then moved three times in eighteen months, each time by an AI system, and each configuration was validated by the verifier of the group that produced it. We re-decide every publicly available witness in exact arithmetic over $\mathbb{Z}[\sqrt{2}]$ on arbitrary-precision integers, sharing no code with any producer, and confirm $K(11) \geq 604$ from four published 604-point configurations. Three, from one platform, have distinct exact contact counts (19,704, 22,904, 22,840), which certifies that they are pairwise non-congruent. The fourth, the headline configuration of the other platform, was unpublished when this ledger first ran: its row measured opacity for 4 days, naming the bytes that would decide it, and decided the day they were published on request. It is congruent to the first of the three: a signed permutation of the coordinates carries one onto the other, and the certificate is verified exactly. The three open rungs of the platform are measured as distances to a witness. No new bound is claimed and no upper bound is touched.

1 The decision procedure

A finite set of non-zero vectors $x_1, \ldots, x_n \in \mathbb{R}^d$ yields a kissing configuration exactly when the pairwise angles are at least $60°$: place a unit sphere at $2x_i/|x_i|$ and all of them touch the central sphere without overlapping. The condition is invariant under scaling each vector separately, so any exact representative of each direction decides the claim, and for $i \neq j$ it reads

$$\langle x_i, x_j \rangle \leq 0 \qquad \text{or} \qquad 4\langle x_i, x_j \rangle^2 \leq \langle x_i, x_i \rangle \langle x_j, x_j \rangle.$$

Both alternatives are polynomial in the coordinates, so a configuration given in any real quadratic field can be decided without error. The 604-point records live in $\mathbb{Q}(\sqrt{2})$, so the certifier works in $\mathbb{Z}[\sqrt{2}]$: a number is a pair (a, b) standing for $a + b\sqrt{2}$ with a, b arbitrary-precision integers, and the sign of $a + b\sqrt{2}$ is decided by comparing a^2 with $2b^2$ when the coefficients disagree in sign.

*Independent researcher. Contact: carlos@carlostoledo.co. This work was produced with substantial AI assistance in a certified-verification setting: constructions and exposition were proposed with the help of large language models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for the certificate to be accepted. The certificates, not the models, are the evidence, and they are public and re-runnable. Machine-derived and not peer-reviewed.

The tie $a^2 = 2b^2$ with $(a, b) \neq (0, 0)$ is impossible over the integers because $\sqrt{2}$ is irrational; the implementation raises on it, and that raise is exercised as a control. No floating-point number participates in any decision.

A decimal input such as -0.1449 is read as the exact rational it denotes, $-1449/10000$, and never as the nearest binary double. This matters in Section 3.

2 Calibration

The instrument is calibrated at every run against two configurations whose kissing numbers are known exactly, generated from their own definitions rather than stored: the D_4 root directions give $K(4) \geq 24$, and the E_8 roots give $K(8) = 240$. The E_8 run reports 6,720 pairs at exactly $60°$, which is the textbook count $240 \cdot 56/2$ for that root system and is reproduced as an equality in exact arithmetic rather than as a float comparison.

3 The dimension-eleven ladder

Theorem 1. $K(11) \geq 604$, *certified in exact arithmetic over* $\mathbb{Z}[\sqrt{2}]$ *from published coordinates, independently of any producer's verifier.*

bound	witness	verdict	exact contacts
582	classical norm-four shell	CERTIFIED	22,560
593	AlphaEvolve, integer coordinates	CERTIFIED	0
594	platform rung winner, decimal coordinates	CERTIFIED	17,088
604	configuration 1, $(a + b\sqrt{2})/6$	CERTIFIED	19,704
604	configuration 2, $(a + b\sqrt{2})/6$	CERTIFIED	22,904
604	configuration 3, $(a + b\sqrt{2})/6$	CERTIFIED	22,840
604	headline claim, $p + q\sqrt{2}$ pairs, published on request	CERTIFIED	19,704
604 in $\mathbb{R}^{12}$	integral D_{12} lift of configuration 3	CERTIFIED	22,840

Each certified row is decided at the shell norm its producer states (four for the three configurations given over six, thirty-six for the headline configuration), and every enclosure is an exact comparison rather than a tolerance.

Corollary 1. *The Station's three* 604*-point configurations are pairwise non-congruent.*

Proof. Congruences preserve the multiset of pairwise inner products and therefore the number of pairs at exactly $60°$. The three configurations have 19,704, 22,904 and 22,840 such pairs, computed as exact equalities in $\mathbb{Z}[\sqrt{2}]$, so no two can be congruent. □

Remark. The rung winner at 594 is instructive about verification rather than about geometry. That platform scores submissions in fixed-precision decimal and switches to exact integer arithmetic only for integer-valued submissions, and the winning vectors are not integral. Read as the exact rationals their decimal literals denote, they are nonetheless a genuine exact witness, with 17,088 pairs at exactly $60°$ and every other pair strictly clear. High-precision decimal is not proof; here it happened to be reporting a true statement.

4 The row that measured opacity, and how it closed

One row of the ledger was not, at first, a geometric verdict. On 2026-09-03 the configuration behind the headline 604 of one platform, which the field's reference tables credit, was not published: the public interface served the solved 594 rung and an open 605 rung, and discussion of the 604 described it without carrying its coordinates. The row read NEEDS DATA, with the threshold stated in one sentence: publish the vectors in any exact or decimal form and the row is decided in minutes.

The ledger asked. On 2026-09-07, 4 days later, the platform's maintainer answered with the repository holding the file: 604 vectors as pairs of integers (p, q) per coordinate, standing for $p+q\sqrt{2}$, every vector at shell norm exactly 36. Pinned by its digest, the file decided CERTIFIED in 65 milliseconds with 19,704 exact contacts. A row of this kind measures the claim-maker rather than the geometry, and the measurement here is favourable: the price of checkability was one request.

5 The two platforms' 604s are one configuration

Theorem 2. *The EinsteinArena* 604*-point configuration and configuration 1 of the Station are congruent. Explicitly, there is a bijection π of the* 604 *vectors and a signed permutation matrix T of the eleven coordinates with $T\,(2a_i) = b_{\pi(i)}$ for every i.*

Proof. Two configurations at one shell norm each are congruent exactly when a bijection matches every pairwise inner product up to the ratio of norms. Each configuration is therefore a complete graph on 604 vertices whose edges carry the exact normalised inner product (22 distinct values here), and congruence is isomorphism of edge-coloured graphs. It was decided by individualisation–refinement: colour refinement as the invariant, backtracking over the smallest cell, the procedure inside NAUTY, implemented independently. The search found π in 11 nodes. From π the matrix T was solved on eleven independent vectors in exact $\mathbb{Q}(\sqrt{2})$ arithmetic and verified on all 604, together with $T^{\mathsf{T}}T = I$; the certificate (π, T) is stored beside the ledger and re-verified at every build without repeating the search. T has entries in $\{0, \pm 1\}$ with one non-zero per row. □

Corollary 2. *Configurations 2 and 3 of the Station are congruent neither to configuration 1 nor to each other. (Their contact counts differ; independently, the same search exhausts at its first node for each pair.)*

Remark. The lineage of the headline configuration is in its bytes. Of its 604 directions, 496 are those of the same platform's 594 rung winner — exactly its 496 integer vectors — and the winner's 98 decimal-valued vectors were replaced by 108 vectors with a $\sqrt{2}$ part; 176 directions are the classical 582 shell's. Its slack is small: 4,608 pairs sit at 60.94°. As data: the EinsteinArena file entered a public repository on 2026-04-12 and its paper was submitted 2026-06-09; the Station's artifacts and paper are dated 2026-08-24.

6 The open rungs, measured

The platform lists three open rungs — $n = 605$ in dimension eleven, $n = 841$ and $n = 842$ in dimension twelve — each with a best submission it scores above zero. The instrument reads

those too, as a distance rather than a verdict: every pair decided exactly, every violation counted, the worst named. The best 605 has 9,510 violating pairs, the worst at 40.24°; the best 842 has 237, the worst at 57.09°; the best 841 fails by exactly one pair, and that pair is a vector repeated: 840 distinct directions with 40,992 exact contacts (41,128 if the repeat is counted), an 840-point configuration submitted as 841. None of this refutes anything: an attempt that fails is not a bound that fails.

7 Scope

No bound is improved here and no configuration was searched for. Upper bounds, which come from semidefinite programming, are untouched; this paper concerns lower-bound witnesses only. Where a producer had already verified its own configuration exactly, the contribution is independence: different code, a different language, different arithmetic, and no shared line with the producer.

Reproduction

Every constant in this paper is interpolated from a certificate record in the `cert-machine` repository, so the document cannot quote a number the machine did not certify. The instrument is `instruments/kissing`; the ledger is `certs/kissing-ledger.json`; the pinned coordinates are in `corpus/kissing` with the upstream digest of each source recorded (the headline 604 file is held byte for byte and re-hashed at every run). Running `node instruments/kissing/battery.js` re-derives the D_4 and E_8 calibrations from their definitions, re-certifies two 604-point configurations live from their pinned bytes, and requires every red control to fire, among them the mixed-sign $\sqrt{2}$ comparator and the exactness of decimal-literal parsing.

References

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